The Layoff Generation: How Generative AI will Reshape Employment and Labor Markets

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Abstract: This research presents a new metric known as "AI Augmentation," aimed at quantifying the influence of generative AI across diverse job roles, organizations, and sectors. The analysis defies prevailing expectations of job losses due to AI, instead demonstrating a reverse correlation between AI Augmentation and layoff rates. These counterintuitive findings highlight the importance of a detailed, multifaceted examination of AI's impact on labor markets. This nuanced understanding is vital for making informed strategic decisions and developing effective policies.

Key Words: Generative AI, AI Augmentation, Layoff Rates, Employment, Labor Markets

Introduction

The transformative potential of large language models (LLMs) like ChatGPT on job markets is gaining significant traction in recent months. Such models, renowned for their capability to generate coherent and contextually relevant text, have swiftly found applications in various economic sectors. Fields like customer support, content creation, and select financial analysis tasks have seen rapid integration of these models (Haleem et al., 2022; OpenAI, 2022; Chang et al., 2023). These advancements echo the arguments of Arntz, Gregory, & Zierahn (2016) and Chui, Manyika, & Miremadi (2016), who posit that while Artificial Intelligence may not eradicate many occupations outright, it is poised to significantly alter the nature of most jobs. Such transformations are contingent on numerous factors, including the nature of the work, the technical feasibility for automation, and the interplay of benefits, costs, and labor dynamics.

However, with the ascent of LLMs, emergent concerns regarding job displacement have arisen. Preliminary data indicates a notable correlation between the incorporation of LLMs and a surge in layoff rates, especially in sectors characterized by routine textual, coding, translate and other repetitive tasks (Chen et al., 2021; Carlini et al., 2021; Felten, 2023; Yilmaz et al., 2023; Jiao et al, 2023; Webb, 2020).

As businesses stand to benefit from reduced operational costs and enhanced efficiency, there emerges a discernible tension between technological progression and job security for certain cohorts of workers. Addressing this transformative phase, our study introduces the "AI Augmentation Index." This tool is designed to gauge the susceptibility of individual professions, companies, and entire industries to LLM influences. It's paramount that decision-makers harness such tools to astutely navigate this evolving landscape, championing a vision where technology complements, rather than supplants, human expertise.

A prevalent assumption is that generative AI, with its potential to automate a plethora of tasks, will invariably lead to widespread job displacement, subsequently creating a 'layoff generation' (Arif et al., 2023). However, these dystopian forecasts often fail to account for the

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intricate dynamics at the interface of AI and labor markets. Understanding the relationship between AI and employment is not a straightforward equation, but rather a complex interplay that unfolds within a multitude of economic, social, and institutional contexts. In this paper, our primary focus centers on the substitutive role of ChatGPT in professional contexts. We assign scores to individual tasks across various jobs. While Eloundou et al. (2023) have categorized tasks, our analysis further incorporates real-world legal and ethical considerations, as well as the interplay between automation and large language models. By harnessing the ChatGPT API for extensive categorization, we ensure that our findings are readily replicable, thereby enhancing the robustness and applicability of our research.

To refine our computation of the AI Augmentation Index, we have taken a two-pronged approach in assessing companies and industries. Firstly, we leverage stock occupation data, akin to the methodology proposed by Eisfeldt et al. (2023), which essentially looks at the total workforce count within a company. Concurrently, in a novel approach, we incorporate data pertaining to the issuance of H-1B visas, highlighting a strong correlation between the number of H-1B visas granted and the employment landscape in the U.S (Mandal, Chaudhuri, & Prasad, 2020). We also consider the monthly increments and decrements of job postings on LinkedIn. The results derived from these diverse data sources not only corroborate each other but also open avenues for us to explore the interplay between these distinct occupational data sets in future research. This dual approach ensures a more holistic and robust understanding of AI's implications across various professional sectors.

In this research, we endeavor to elucidate the relationship between the AI Augmentation Index and employment patterns. Drawing on the work by Fingleton, Garretsen, and Martin (2012), as part of our regression construction, we've incorporated both event shocks and regional factors as control variables, ensuring a more nuanced understanding of the connection between AI Augmentation and layoff rates. Based on current labor market dynamics, our overarching conclusion indicates that companies and industries with higher AI Augmentation tend to exhibit lower layoff rates. However, in specific sectors, a contrasting trend emerges, where increased AI Augmentation correlates with higher layoffs. This counters the prevailing narrative suggesting a looming 'layoff generation' due to generative AI. Such a binary perspective on AI's impact might risk obscuring the intricate interplay between technology and employment, potentially hindering strategic decision-making and policy formulation. Additionally, our results indirectly affirm the perspectives of Zarifhonarvar (2023), Nah et al. (2023), and Nakamura and Zeira (2018). Given these findings, we advocate for a transition in discourse, moving from overarching concerns of massive layoffs to a more detailed examination of the multifaceted relationship between AI and the job market.

This research intends to encourage stakeholders across academia, industry, and policy-making to transition from a perspective of anxiety and uncertainty to one of informed understanding and strategic action. After all, the advent of generative AI presents not only challenges but also opportunities - opportunities for job creation, industry evolution, workforce upskilling, and overall socio-economic development, which must be leveraged in the era of AI-driven transformations.

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Literature review

The pervasive influence of Large Language Models (LLMs) on the labor market is profound and unpredictable, particularly concerning labor demand. (Korinek and Stiglitz, 2018; Bessen, 2018; Acemoglu et al., 2020; Benzell et al., 2021) With the introduction of ChatGPT, AI's transformative potential is reshaping modern industries, innovatively blending technologies like machine learning, natural language processing, and robotics. This has facilitated AI's seamless assimilation into numerous sectors, setting the stage for a new economic landscape.(Aghion et al., 2018;) This literature review seeks to offer a thorough exploration of the AI exposure across S&P 500 companies, shedding light on AI's multifaceted impact, and unearthing potential avenues for further investigation.

(Brougham and Haar, 2018) delve into employees' conceptualizations of AI's role within professional ecosystems. They elucidate that discerning tasks most prone to automation may unveil avenues to recalibrate employee engagement and optimize digital labor platforms. Different from automatic technologies, Generative artificial intelligence (AI) is able to perform cognitive functions and engage in creative endeavors. The application of generative AI would dramatically reshape the fabric of work as we comprehend it. Firms are able to deconstruct occupations into task bundles and assigned different tasks to narrow AI and generative AI based on the existing algorithms (Castillo, 2023). Such insights underscore the imperative to quantitatively assess AI's penetration, paving the way for its more nuanced assimilation in the workplace.

Intricately extending this concept, our research bifurcates AI Augmentation into two distinct facets: augmentation and replacement. This dichotomy allows us to offer a more granular perspective on the varying impacts of AI across different occupations. The rationale for this bifurcation stems from the understanding that the effects of AI on jobs are not homogeneous. Instead, these effects are contingent on whether AI technologies augment human efforts by assuming certain tasks or supplant humans entirely in specific job roles. The concepts of augmentation and replacement form a critical dichotomy in the discussion of AI's influence on employment. Augmentation represents a scenario in which AI technologies serve as an enabler, enhancing the efficiency of individuals in performing specific tasks. Conversely, replacement represents a scenario where AI technologies assume the entire role of a given position, effectively eliminating the need for human involvement. This latter scenario can be viewed as an extreme manifestation of augmentation. The transition from augmentation to replacement is not seamless, however, and is subject to three primary constraints. By distinguishing between augmentation and replacement, we offer a more comprehensive framework for analyzing AI's labor market impact.

While distinguishing between augmentation and replacement, we consider automation to be the key component in the framework. Our research contributes a novel angle to the discourse on whether automation complements or replaces labor, a question that has been considered but not fully answered in the existing literature. (Lawrence et al., 2017) postulate that automation may lead to job losses, and the extent of these losses depends on whether automation supplements or replaces labor. Similarly, (Eloundou et al., 2023) present two key concepts to explain the labor market impacts of Large Language Models (LLMs). They discuss the skill-biased concept of technological change, which posits that technological advances favor more skilled and educated workers, and the automated task model, which

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suggests that generative AI is used to automate specific tasks rather than entire jobs. (Noy and Zhang, 2023) provides an answer to the question of how generative AI systems affect employee productivity in the context of intermediate-level professional tasks, in particular the aspect of whether they replace the employee's efforts or complement the employee's skills.

Taking into account ethical and legal considerations, as well as the current limitations of generative AI technologies, we have incorporated these factors into our framework when formulating the rubric for calculating the AI Augmentation index. For instance, legal ambiguities surrounding liability may restrict the application of generative AI tools such as ChatGPT in sensitive fields like healthcare (Dave, Athaluri, & Singh, 2023). (Jiang et al., 2021) indicate that amidst the rapid development and integration of AI in the medical domain, several emerging challenges have surfaced, especially the lack of standardized processes, ethical and legal concerns, and other impediments . Furthermore, technological deficiencies in integrating large language models with automation could inhibit the replacement of certain occupations (Ahmad et al., 2023). This limitation is compounded by the current stage of large language model technology development. In handling intricate problems, large language models often display a 'hallucination effect', whereby they fabricate terms or facts as needed (Shen et al., 2023). This highlights an existing technical shortcoming in the evolution of large language models.The importance of differentiating between full job replacement and augmentation of human capabilities by AI is underscored by these complexities, and our paper makes a significant contribution by providing a nuanced analysis of these distinct scenarios.

(Eisfeldt et al., 2023) illuminates the influence of Generative AI on firm values through a unique firm-level exposure metric. This marks an evolution from prior research that primarily considered occupational exposure. The authors' innovative approach, which maps the aggregate of task-level exposure metrics to the occupations in the O*NET database (O*NET, 2023), and then to publicly traded companies, provides a new lens through which to explore the economic repercussions of AI advancements. This method demonstrated that firms with higher exposure to Generative AI technologies experienced higher excess returns following the release of ChatGPT.

We consider the analysis of the labor market in the context of evaluating the economic impacts of generative AI is of paramount importance, as it directly influences productivity, income distribution, consumer demand, and structural shifts in the economy. Generative AI technologies can potentially automate routine tasks and amplify efficiency, thereby impacting productivity and economic growth rates. (Agrawal et al., 2019) against drawing broad conclusions from factory automation studies, emphasizing the need for further research into the effects of AI on the labor market . Therefore, a detailed labor market analysis can provide critical insights for informing strategies to maximize the benefits of AI.

Building upon this groundwork, our research brings an original perspective to the field by offering a more comprehensive, nuanced analysis that leverages diverse datasets, including H-1B visa records, job postings trends, and data from Revelio Labs. Our method disrupts the pervasive belief that AI will invariably cause widespread unemployment and instead presents a deeper understanding of the intricate relationship between technology innovation and labor market dynamics. We extend our analysis to all 25 industries in the S&P 500 Index, correlating AI Augmentation scores with layoff trends for each industry's

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companies. This unique and meticulous approach uncovers nuanced patterns, substantiating the hypothesis that Generative AI impacts labor markets in multifaceted and surprising ways. Notably, our work underscores that the effects of Generative AI on employment are far from uniform, varying significantly across different industries. The groundbreaking insights we provide will significantly contribute to the ongoing discourse on the socio-economic impact of AI.

While (Eloundou et al., 2023) acknowledges that generative AI is changing the labor market by providing new job opportunities while AI advancement also displacing some occupation, they caution that it is too early to definitively predict the potential impact of LLMs on the U.S. labor market.

In contrast to this cautionary stance, our research provides concrete evidence of generative AI's impact on the labor market in the short term. It even suggests that these effects might be predictable in the long term. We demonstrate that some industries are already witnessing a rise in unemployment rates while others are experiencing the contrary. This aligns with Eloundou's premise but significantly extends it by providing tangible, real-world data that illustrates the nuanced and differential impacts of generative AI across industries. Our findings, therefore, are not just complementary to Eloundou's theoretical framework but also serve as a valuable empirical extension of it. We have uncovered industry-specific patterns that further elucidate the relationship between AI Augmentation and employment trends, thereby contributing robust evidence to the ongoing discourse on the socio-economic implications of AI technology.

Our research unveils that the relationship between AI Augmentation and unemployment rates is particularly evident in certain industries. Notably, the Automobiles and Components, Software and Services, Semiconductors and Semiconductor Equipment, and Consumer Services sectors show a strong correlation between higher AI Augmentation and increased unemployment rates. These are the very industries that are currently spearheading the ongoing technological revolution, strategically employing advanced technologies to spur innovation, enhance efficiency, and create unprecedented products or services. Taking the automobile industry as a case in point, we are witnessing a remarkable transformation catalyzed by the advent of electric vehicles and autonomous driving technologies. These advancements drastically reshape the transportation landscape, and our research provides critical insights into their labor market implications. Similarly, in the retail sector, AI and data analytics integration are revolutionizing operations, enabling personalized marketing strategies and more efficient inventory management. Our findings underscore the labor market shifts induced by such technological integration. In the realm of biotechnology and pharmaceuticals, the fusion of genomics and AI is expediting the drug discovery process, marking the dawn of personalized medicine. Meanwhile, the transportation and utilities sectors are capitalizing on technological advancements, optimizing logistics and energy distribution through innovative grid technologies and automation.

On the other hand, we find that for certain industries, including Banks, Equity Real Estate Investment Trusts (REITs), Health Care Equipment and Services, Household and Personal Products, Insurance, and Media and Entertainment, the effect of generative AI appears to be optimistic. These sectors do not exhibit the short-term negative impacts seen in industries with higher technical barriers. Jobs in these industries, such as retail salespeople,

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warehouse clerks, and caregivers, typically involve many repetitive tasks, data collection, and basic data processing. These tasks can be efficiently managed with automation tools, while aspects requiring human interaction and physical manipulation remain irreplaceable. (Nah et al., 2023) proposes that, particularly in industries such as healthcare, ethical considerations must be considered. They believe that the development of generative AI applications requires careful examination of ethical and regulatory issues, as well as the necessity for collaboration between artificial intelligence and healthcare professionals. Unlike the technology industry, which requires a significant period to adapt to the technological innovations introduced by raw AI, the jobs mentioned do not require extensive time to acquire the knowledge and skills needed to handle automated systems. This results in a paradoxical trend where higher AI Augmentation corresponds to lower unemployment rates.

(Frank, et al., 2019) noted the uncertain impact of technology on labor makes it challenging to construct resilient labor markets. Technological changes can cause shifts in specific skill demands, leading to broader labor trends, such as the redefinition of occupational skills, employment redistribution, and worker migration. Similarly, according to findings by (Frey and Osbone, 2016), as technology advances, low-skill workers will need to transition to tasks impervious to computerization, specifically those requiring creative and social intelligence. Thus, for workers to maintain competitiveness, the acquisition and enhancement of these skills are imperative.

In essence, our work exposes the nuanced and industry-specific impacts on employment. It unravels the complex labor market dynamics in these sectors, elucidates the multifaceted impacts of AI, and offers a more granulated understanding of the socio-economic ramifications of AI adoption across different industries.

As discussed above, our research presents compelling evidence that the impacts of generative AI on labor markets are primarily short-term. This observation aligns with (Zarifhonarvar, 2023) theory , which posits a short-term disruption followed by potential long-term equilibrium. The short-term effects we have documented lend empirical weight to this theory while paving the way for a more nuanced understanding of the long-term implications of AI on job markets. This short-term disruption manifests in the form of a 'skills mismatch.' As generative AI technologies like ChatGPT replace certain job roles, there is a disjunction between the skills demanded by the new, AI-integrated job market and the capabilities of displaced employees. This mismatch can result in prolonged unemployment for those unable to re-skill or transition to new roles. Although opportunities for re-skilling may mitigate the adverse effects of AI on the job market, the reality of time constraints and the fast pace of AI adoption can lead to temporary mismatches between worker skills and employer needs. This, in turn, may result in unemployment or underemployment in specific sectors.

Our research, therefore, substantiates above theoretical predictions and contributes novel, empirical insights into the differential impacts of AI Augmentation across various industries. This enriches our understanding of the short-term disruptions and long-term implications of AI on labor markets, while (Bessen, 2019) also posits that technologies enhancing productivity may augment employment within certain sectors. Utilizing data spanning two centuries from the manufacturing industries, the Bessen demonstrates a stimulatory effect of automation on employment, which increased productivity and job proliferation. Bessen

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speculates that contemporary technology could precipitate contraction in some industries while fostering growth in others. Consequently, while automation may not directly engender mass unemployment, it could necessitate disruptive labor transitions towards burgeoning industries, thereby demanding the acquisition of novel skills and occupations.

We assume an optimistic outlook on the long-term impact of generative AI on the job market, a perspective substantiated by the following works. (Nakamura and Zeira, 2018) constructed a task-based theoretical model, and the results are relatively optimistic. By analyzing the mathematical modeling and analysis of this model, the model suggests that automation does not necessarily lead to unemployment and that unemployment due to automation is expected to decline over time. (Lim and Lee, 2022) provides a unique insight into how technological advancement affects employment, discussing the relationship between innovation and employment, with a focus on the impact of product and process innovation on job creation . For the purposes of this paper, process innovation refers to the introduction of new or improved methods of production, such as changes in the way goods or services are produced, delivered, or distributed. Product innovation, on the other hand, refers to the introduction of new or improved goods or services, such as new or improved products, services, or processes. Lim and Lee conducted a firm-level empirical analysis using long-term data on Korean firms and industries, confirming that product innovations have a positive employment impact, while the impact of process innovations tends to be varied or negligible. Product innovation has a positive impact on employment growth in the long run, and even though product innovation tends to have a negative impact in the short run, this negative impact will be offset by positive spillovers in the long run. In the long run, however, generative AI has positive externalities, such as increased productivity and innovation efficiency, which can lead to companies developing and improving products at a faster rate, thus creating more jobs, and increasing employment.

(Autor, 2015) contends that while automation has indeed been developed to substitute labor, it concurrently serves as a complement to labor, augmenting output and consequently catalyzing an increased demand for labor. This nuanced perspective challenges prevalent narratives that overstate the displacement of human labor by machines neglecting the symbiotic relationship between automation and labor that bolsters productivity elevates earnings, and fuels labor demand. In considering the advent of artificial intelligence and robotics, the author suggests that the interplay between machine and human comparative advantage facilitates the substitution of workers in routine tasks whilst simultaneously amplifying the human advantage in problem-solving, adaptability, and creativity.

(Lawrence, Roberts and King, 2017) also share the same view with (Lim and Lee,2022) , which indicated that AI automation has a positive for the job market because it has a substantial positive spillover effect, which in the long run, AI automation has a positive impact on the job market. In the long run, AI automation may change the way work is done and productivity rather than eliminating jobs. Lawrence argues that automation may lead to job losses in the short term, but in the long run, it may also create new jobs. Also, the paper suggests that if demand keeps pace with the increase in output, employment levels may remain constant or even grow. But it will take businesses as well as society to decide how automation will reshape the economy as well as the industrial landscape.

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Data and Methodology

The Steps Involved in Calculating AI Augmentation

In this study, we propose a systematic approach to quantify an organization's Augmentation to artificial intelligence (AI), comprised of four methodical steps. The process is illustrated in Figure 1. We base our computation approach on the groundwork provided by (Eisfeldt et al., 2023), enhancing it with our novel contributions.

Initially, we employ the O*NET database (version 27.3) to obtain comprehensive information on the tasks involved in a variety of occupations (Tolan et al., 2021). With the aid of ChatGPT, we generate an AI Augmentation index for each profession, which gauges the likelihood of these occupations being affected by AI technologies. The second step involves harnessing data released by the United States Citizenship and Immigration Services (USCIS), specifically focusing on the annual count of H-1B visas filed by firms for different roles. We introduce a weighting scheme that considers the diversity of occupations to derive an H-1B based AI Augmentation index for each firm. In the third phase, we meticulously gather and process data from LinkedIn, keeping track of the frequency of job postings published and subsequently withdrawn by firms on a monthly basis. This allows us to calculate a 'post and remove' based AI Augmentation index for each firm, indicating the company's potential engagement with AI. Finally, we integrate the data obtained from the second and third steps, applying a weighted integration approach to compute a refined, firm-level AI Augmentation index. This holistic measure is meticulously designed to ensure utmost precision and data integrity in our estimation of a firm's AI Augmentation.

Notably, the 'H-1B' and 'post and remove' measures we employ are based on incremental occupational data, capturing changes in the job market. In parallel, we also reference findings from Revelio Labs, which uses a different approach. Revelio Labs capitalizes on stock occupational data, which represents the accumulated job positions within a company, to calculate firm Augmentation. In this study, we employ a cross-utilization strategy, integrating the results from our incremental data with Revelio Lab's stock occupations data to provide a comprehensive perspective on firm-level AI Augmentation.

Rubric of Replacement

Drawing upon the methodologies of (Brynjolfsson, 2018; Felten et al., 2018; Webb, 2020), we have likewise developed a standardized measure for the tasks associated with each occupation. We introduce a comprehensive task-based classification rubric that evaluates the potential of AI to replace tasks across various occupations. It offers a nuanced assessment of the degree of AI-replacement potential, considering factors beyond mere technical feasibility, and encompassing legal, ethical, and social dimensions. The rubric is divided into three categories: No Augmentation (A0), Weak Augmentation (A1), and Strong Augmentation (A2), with detailed criteria outlined for each category. A detailed representation of the rubric can be found in Table 1.

The "No Augmentation" category (A0) includes tasks that are currently not suitable for automation or AI augmentation, often due to the high level of judgement, interaction, and adaptability required. These tasks often deal with complex human interactions, unpredictable situations, or involve important decision-making that can significantly affect human lives or livelihoods. Moreover, they may also involve activities that legally require human

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involvement.

The "Weak Augmentation " category (A1) includes tasks that can be partially augmented by technology. These tasks are typically characterized by a blend of structured, automatable tasks and unstructured tasks that necessitate human judgment or interaction. In many of these tasks, technology can assist or augment human performance, but not entirely replace it. These tasks can also require regular oversight, adjustment, or input from a human.

Lastly, the "Strong Augmentation" category (A2) includes tasks that can be fully augmented by technology. The tasks involved are typically highly structured and repeatable, without a necessity for human judgement, creativity, or complex human interaction. Automation can perform these tasks consistently and at scale, and they do not involve making decisions that might significantly impact human lives or livelihoods. Importantly, these tasks do not legally require human involvement.

This rubric builds upon and innovates beyond the foundation laid by Eloundou et al. (2023), offering several notable advantages. First, it integrates ethical, legal, and social considerations, making it more comprehensive and holistic. Second, by discerning between tasks that can be fully, partially, or not at all replaced by AI, it provides a more fine-grained understanding of AI's potential impact on the workforce. Thus, this approach equips us with a robust and nuanced tool for forecasting the transformative effects of AI on various occupations.

AI Augmentation in S&P 500

Occupation level AI Augmentation

By applying the procedural framework described earlier, we have compiled AI Augmentation indices for 923 unique occupations. To portray the broad range of AI Augmentation inherent in these positions, we have opted to spotlight the occupations at both ends of the spectrum – the top 10 and bottom 10 – ranked by their respective AI Augmentation. These are presented for ease of reference in Table 2.

Firm level AI Augmentation

Through the computed AI Augmentation for each occupation, we are able to derive a firm's AI Augmentation based on the composition of its workforce. Utilizing the stock occupation data from Revelio Labs, we have calculated the AI Augmentation for each company within the S&P 500. Figures 2 and 3 depict the AI Augmentation in the top and bottom ten companies by market capitalization within the S&P 500, specifically illustrating the three most abundant occupations within these firms.

Our calculations of AI Augmentation for both stock and incremental occupations have been sorted, and the top ten companies with the highest AI Augmentation are listed in Table 3 and Table 4, respectively. These tables provide a comprehensive view of the industries most affected by AI Augmentation, helping stakeholders to better understand the current AI trends and potential future shifts in the job market.

Industry level AI Augmentation

Drawing from the cumulative AI Augmentation indices developed using H-1B visa and LinkedIn 'post and remove' data, indicative of an incremental perspective, as well as the

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Revelio Labs dataset, representing a stock viewpoint, we undertook a thorough examination of all constituent corporations within the S&P 500. For the purpose of sectoral stratification, we employed the Global Industry Classification Standard (GICS), thereby facilitating a segmented analysis. The subsequent computation of AI Augmentation at an industry level was realized by determining the mean AI Augmentation of firms within each respective industry, with the outcomes visually represented in Figure 4. Similarly, we have also ranked each industry within the S&P 500 index based on both stock and incremental data. The respective tables are presented as Table 5 and Table 6.

The Relationship between Generative AI and Employment

Upon establishing the AI Augmentation index for each company, we further supplemented this analysis with information on the layoff rates and total numbers of layoffs for each corporation in the S&P 500, leveraging data procured from Revelio Labs spanning the years 2019 to 2023. This encompassing time frame allows us to inspect potential temporal trends and the influence of external events, such as the advent of COVID-19, on AI Augmentation and subsequent layoff patterns. The relationship between unemployment rates and AI Augmentation is illustrated in Figure 5. In addition, to offer a more comprehensive exploration of the dynamics between AI Augmentation and unemployment rates, we formulated a regression model. This approach allows us to delve deeper into the intricate correlations between these two elements.

$$R_{\text{Job Layoffs}} = \alpha_{job} Augmentation_{job} + \alpha_{\text{firm}} Augmentation_{\text{firm}} + \alpha_{\text{gpt}} \mathbb{1}[\text{GPT impact}] + \alpha_{\text{Covid}} \mathbb{1}[\text{Covid impact}] + \sum \alpha_{\text{states}} \mathbb{1}_{\text{States}} \quad (1)$$

The layoff rates ($R_{\text{Job Layoffs}}$) serve as our dependent variable, reflecting the rate of job terminations for the companie's occupations under study. The variable $\text{Augmentation}_{\text{job}}$ measures the generative AI Augmentation at the occupational level within these firms, while $\text{Augmentation}_{\text{firm}}$ provides a comprehensive assessment of the firm's total AI Augmentation. These two variables are the focal point of our investigation, expected to shed light on how susceptibility to AI influences labor dynamics within these organizations.

Accounting for temporal factors, we include GPT impact and Covid impact as time dummy variables in our model. GPT impact is constructed to capture the potential effect of generative AI models, especially since the advent of GPT-3 in June 2020 and GPT-4 in 2022. Covid impact, on the other hand, seeks to capture the labor market effects associated with the COVID-19 pandemic, an extraneous variable that significantly altered the business landscape during our period of study.

Lastly, to control for regional influences, we introduce the States variable, which segments the companies into four geographic categories based on their primary hiring locations. The categories include the East Coast, Midwest, West Coast, and Other regions. This variable enables us to account for any potential regional disparities in the labor market dynamics or technology adoption patterns.

The results of this multivariate regression analysis are provided in Table 7. This comprehensive representation allows us to discern the degree of influence each variable has on the dependent variable, the layoff rates. By interpreting these results, we aim to enhance our understanding of the intricate relationship between a firm's AI Augmentation and its employee layoff rates, in the context of major U.S. corporations within the S&P 500.

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Following the comprehensive regression analysis for all S&P 500 companies, we further drilled down to perform regression analyses by industry within the S&P 500. The coefficients for each factor across different industries were visualized in a bar chart (Figure 6). This graphical representation offers a clearer understanding of how AI Augmentation, the introduction of GPT technologies, and the COVID-19 pandemic influenced layoff rates across various industries. The diversity in the impacts depicted underscores the importance of considering industry-specific contexts when investigating the effects of these factors on employment patterns.

In addition to our sector-based analysis, we further present a company-specific exploration. We have identified the top 20 companies with the highest layoff rates and total number of layoffs in 2023, contrasting these figures with their AI Augmentation levels from 2022. These relationships are graphically depicted in Figures 7 and 8. Similarly, we have determined the top 20 companies with the highest recruitment rates and total number of hires, presenting these alongside their 2022 AI Augmentation levels in Figures 9 and 10.

Both occupation and firm-level AI Augmentation are inversely correlated with layoff rates, suggesting that higher AI Augmentation may be associated with lower layoffs, potentially due to effective adaptation to market shifts. The introduction of GPT technologies appears to correlate with decreased layoffs, possibly reflecting AI-induced efficiency gains. In contrast, the onset of the COVID-19 pandemic marked a rise in layoffs, likely due to wide-scale economic disruptions. Regional differences also emerge, with the East Coast, Midwest, and West Coast showing positive associations with layoff rates, possibly indicating varying regional economic conditions or industry compositions.

Moreover, it's worth noting that these results are reflective of the current state of affairs, and the full impact of generative AI on employment may not yet have manifested. The displacement effect of generative AI may still be in its infancy, and the long-term consequences remain to be seen. The transformation brought by generative AI is profound, necessitating continuous observation and analysis as it evolves over time.

Conclusion

Our research reveals that the discourse surrounding the advent of AI and its impact on labor markets is more complex than widely perceived. While the introduction of generative AI into various sectors does present a potential for job displacement, our study demonstrates an inverse relationship between AI Augmentation and layoff rates. This finding challenges the common notion of an impending "layoff generation" due to AI. The complexities inherent in the relationship between AI and employment necessitate a shift in narrative and a more nuanced approach to strategic decision-making and policy formulation.

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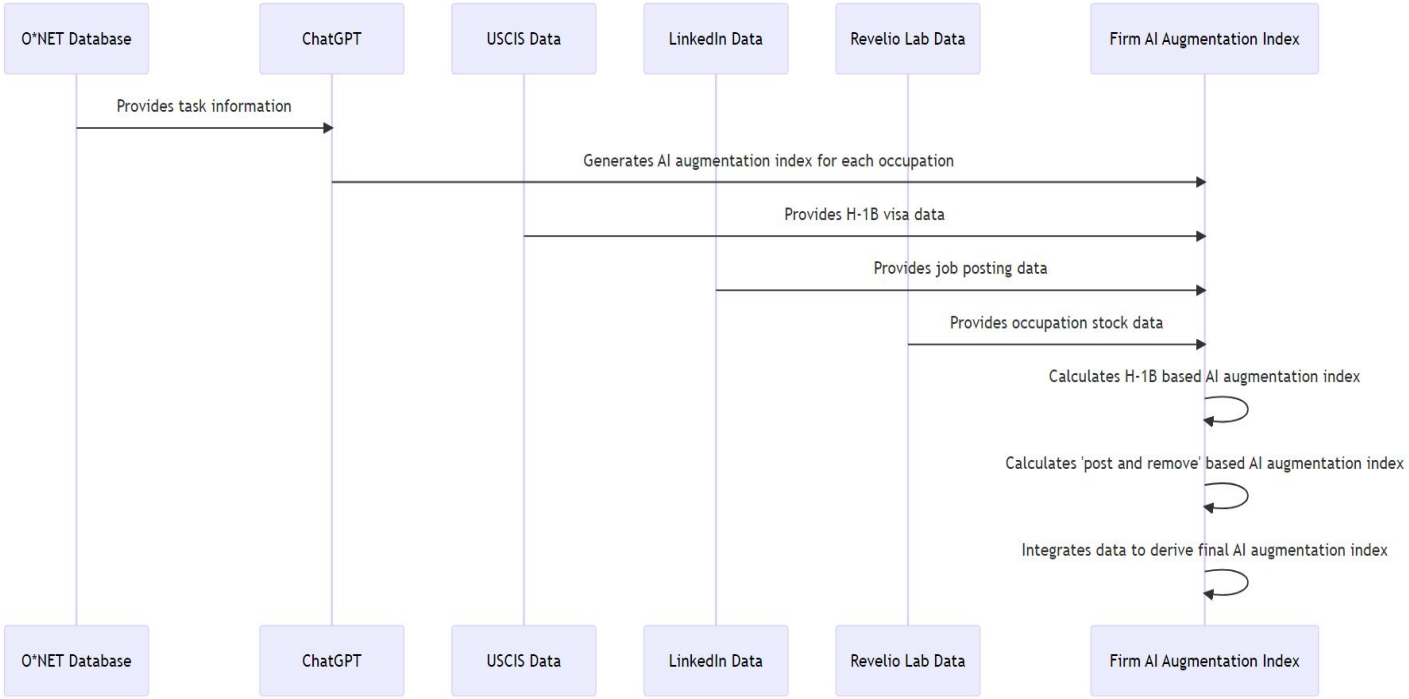
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Appendix

Figure 1: The Steps Involved in Calculating AI Augmentation



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Table 1: Replacement Rubric

Category	Criteria
A0 - No Augmentation	1. The task requires high degree of human judgment or decision-making based on complex and non-standardized criteria, which the current technology cannot reliably support. 2. The task requires a significant degree of human interaction or personalized service where a "human touch" is critical. 3. The task involves physical manipulation or navigation in a complex, unpredictable environment that technology cannot adapt to or operate in effectively. 4. The task requires a deep understanding of, and ability to navigate, complex human social and cultural norms that are beyond the scope of current technology. 5. The task involves dealing with unexpected situations or emergencies that require human adaptability and critical thinking skills. 6. The task involves decisions or activities that significantly impact human lives, and which therefore carry a high level of responsibility and accountability. 7. Legal restrictions or ethical considerations prevent the application of technology to assist or augment the task.
A1 - Weak Augmentation	1. The task involves a combination of structured elements that can be supported by technology and unstructured elements that require human judgment. 2. The task occurs in a variable context or environment that requires human adaptability, even though certain aspects of the task can be assisted by technology. 3. Technology can enhance human performance in the task, but cannot fully replace the human due to the need for human judgment, creativity, or personalized service. 4. The task involves processing complex information where technology can assist in preliminary analysis, but final interpretation or decision-making requires human expertise. 5. The task requires physical interaction in a partially structured environment, where technology can assist but not fully replace human involvement. 6. The task requires regular human oversight, adjustment, or input to ensure correct performance, even though certain aspects can be automated. 7. Legal or ethical considerations allow only for partial technological augmentation of the task.
A2 -Strong Augmentation	1. The task involves highly structured, repeatable activities that can be significantly enhanced by technology. 2. The task can be completed through digital or virtual means, or involves physical activity in a highly controlled and predictable environment, making it conducive to technological augmentation. 3. The task involves processing or analyzing structured data where automation tools, including LLM, can perform as well or better than humans. 4. Technology can perform the task consistently and at scale, reducing the need for human oversight or intervention. 5. The task does not involve complex human interaction or require understanding of nuanced social or cultural norms. 6. The task does not involve decisions that would significantly impact human lives or livelihoods, making it safer and more ethically acceptable for technology to take on a larger role. 7. There are no legal restrictions or ethical considerations preventing the use of technology to augment the task to a significant degree.

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Table 2: Partial Occupational AI Augmentation

Rank	Occupation	SOC_Code	Occupation AI Augmentation
1	Bookkeeping, Accounting, and Auditing Clerks	43-3031.00	0.6722222
2	Climate Change Policy Analysts	19-2041.01	0.6309524
3	Secretaries and Administrative Assistants, Except Legal, Medical, and Executive	43-6014.00	0.6197917
4	Computer Programmers	15-1251.00	0.6078431
5	Data Scientists	15-2051.00	0.6041667
6	Tax Preparers	13-2082.00	0.5972222
7	Writers and Authors	27-3043.00	0.5916667
8	Computer Network Support Specialists	15-1231.00	0.5769231
9	Budget Analysts	13-2031.00	0.5769231
10	Bioinformatics Technicians	15-2099.01	0.5701754
...	...	...	...
914	Motion Picture Projectionists	39-3021.00	0.0000000
915	Motorcycle Mechanics	49-3052.00	0.0000000
916	Occupational Therapy Aides	31-2012.00	0.0000000
917	Oral and Maxillofacial Surgeons	29-1022.00	0.0000000
918	Orthodontists	29-1023.00	0.0000000
919	Physical Medicine and Rehabilitation Physicians	29-1229.04	0.0000000
920	Physical Therapist Assistants	31-2021.00	0.0000000
921	Reinforcing Iron and Rebar Workers	47-2171.00	0.0000000
922	Sewers, Hand	51-6051.00	0.0000000
923	Surgical Assistants	29-9093.00	0.0000000

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Figure 2: Top 10 Companies by AI Augmentation Among the Top 100 Market Cap Firms in the S&P 500, 2023

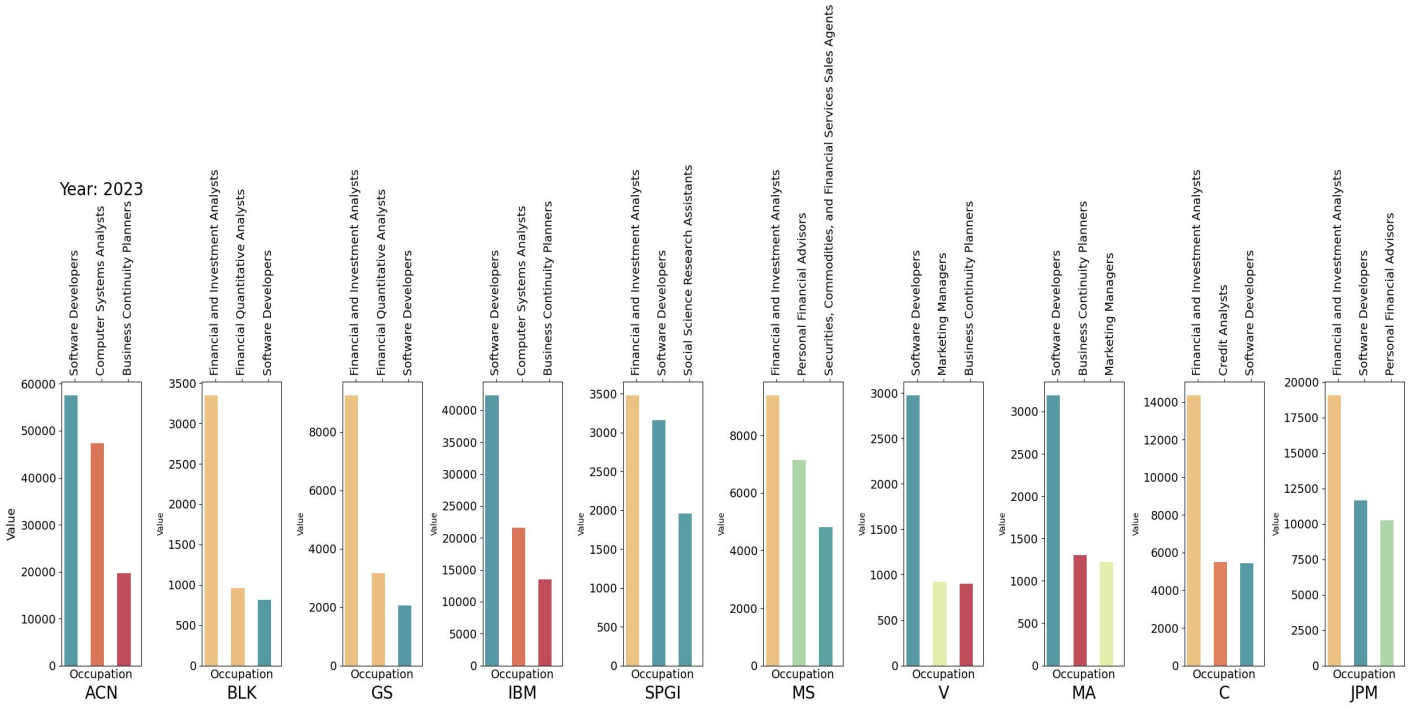
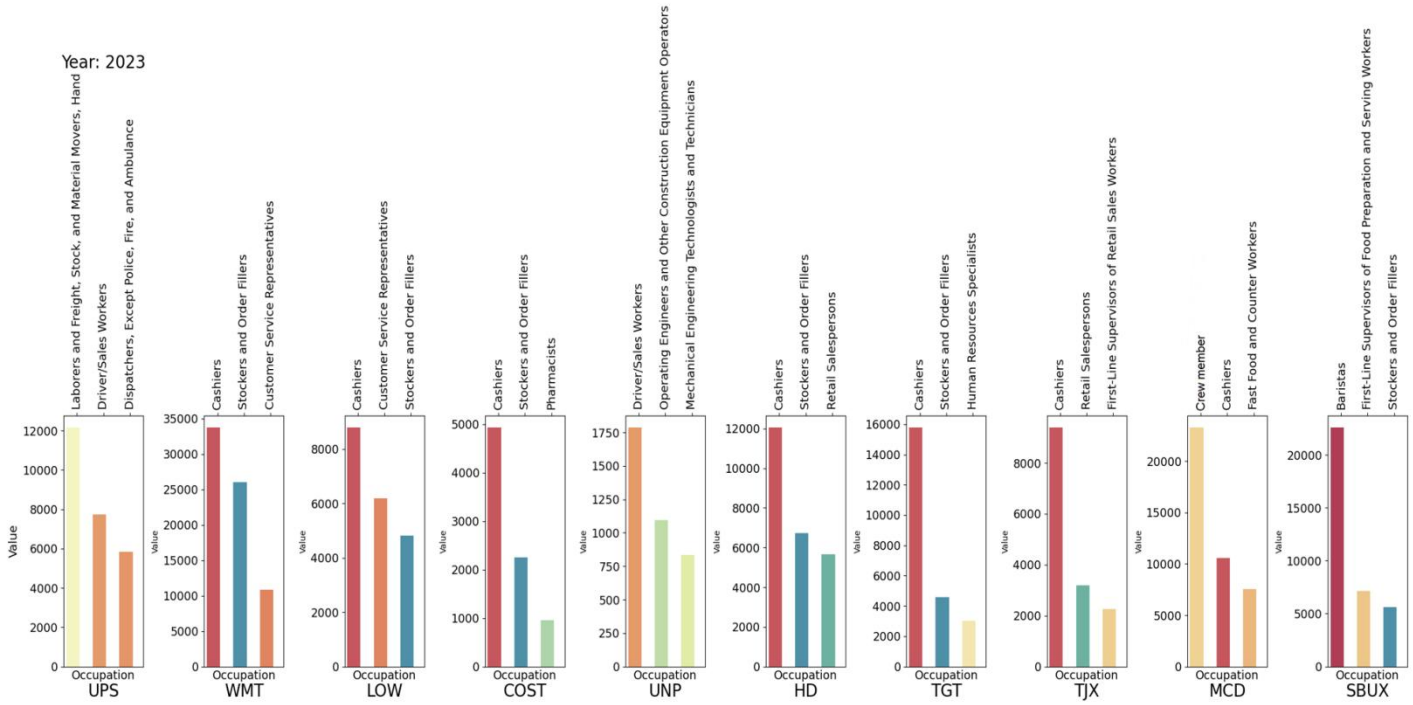


Figure 3: Bottom 10 Companies by AI Augmentation Among the Top 100 Market Capitalization Firms in the S&P 500, 2023



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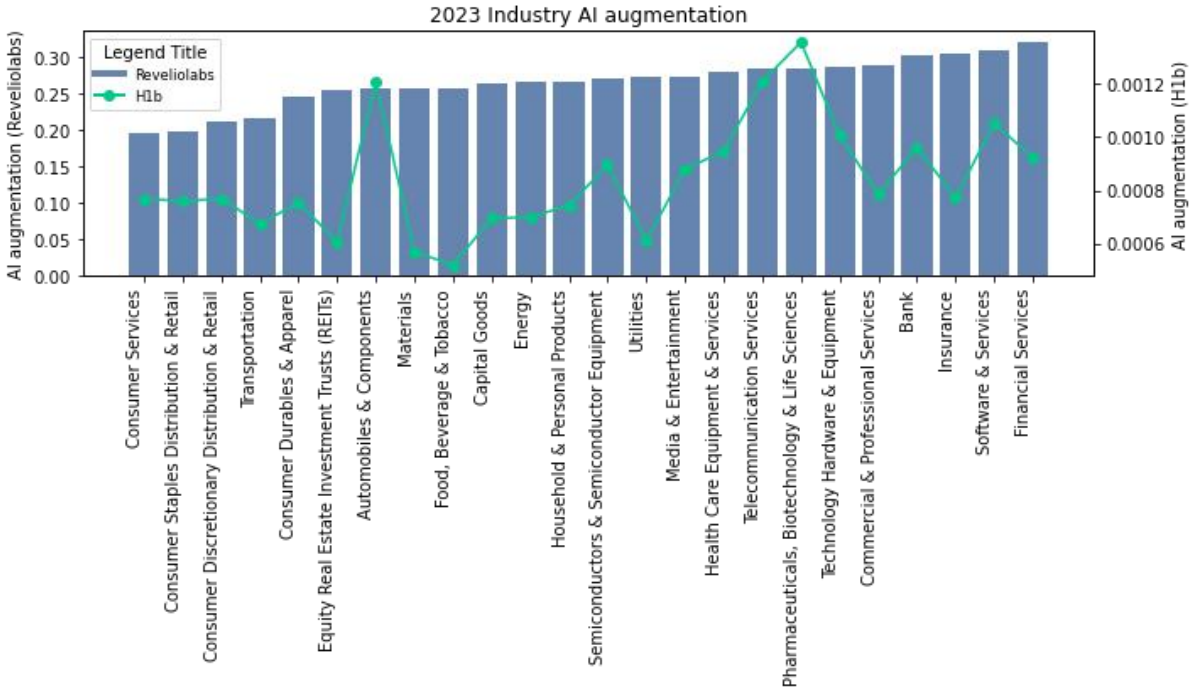
Table 3: Top 10 Companies with the Highest AI Augmentation Based on Stock Data

ticker	Industry	2019	2020	2021	2022	2023
ACN	Software & Services	1	2	2	3	2
RE	Insurance	2	1	1	1	1
NTRS	Financial Services	3	3	6	6	5
STT	Financial Services	4	4	7	8	10
BLK	Financial Services	5	5	3	4	9
BK	Financial Services	6	8	10	5	8
DXC	Software & Services	7	10	9	9	6
ACGL	Insurance	8	6	5	2	4
MSCI	Financial Services	9	9	4	10	7
CTSH	Software & Services	10	7	8	7	3

Table 4: Top 10 Companies with the Highest AI Augmentation Based on Incremental Data

Ticker	Industry	2020	2021	2022	2023
C	Banks	1	1	1	2
ACN	Software & Services	2	2	4	1
MSCI	Financial Services	3	3	3	8
IBM	Software & Services	4	8	5	6
DXC	Software & Services	5	11	2	3
FITB	Banks	6	5	17	30
BK	Financial Services	7	13	6	78
MS	Financial Services	8	12	20	33
GPN	Financial Services	9	57	49	139
IVZ	Financial Services	10	4	12	4

Figure 4: 2023 Industry AI Augmentation



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Table 5: Ranking of Industries within the S&P 500 Based on AI Augmentation from Stock Data

Industry	**2019**	**2020**	**2021**	**2022**	**2023**
Pharmaceuticals, Biotechnology & Life Sciences	1	1	1	1	1
Health Care Equipment & Services	2	2	2	2	2
Utilities	3	3	3	3	3
Materials	4	4	4	4	4
Insurance	5	6	5	5	5
Financial Services	6	5	6	6	6
Technology Hardware & Equipment	8	7	7	7	7
Energy	7	8	9	8	8
Software & Services	10	9	8	11	9
Household & Personal Products	13	12	12	10	10
Semiconductors & Semiconductor Equipment	9	10	10	9	11
Capital Goods	11	11	11	12	12
Commercial & Professional Services	12	13	13	13	13
Food, Beverage & Tobacco	14	14	14	14	14
Bank	17	16	15	15	15
Automobiles & Components	15	15	16	16	16
Telecommunication Services	18	17	18	18	17
Real Estate Management & Development	16	18	17	17	18
Media & Entertainment	19	19	19	19	19
Equity Real Estate Investment Trusts (REITs)	20	20	20	20	20
Consumer Durables & Apparel	21	21	21	21	21
Transportation	22	22	22	22	22
Consumer Services	23	23	23	23	23
Consumer Discretionary Distribution & Retail	24	24	24	24	24
Consumer Staples Distribution & Retail	25	25	25	25	25

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Table 6: Ranking of Industries within the S&P 500 Based on AI Augmentation from Incremental Data

Industry	**2020**	**2021**	**2022**	**2023**
Banks	1	1	1	3
Real Estate Management & Development	2	2	2	1
Software & Services	3	3	4	5
Telecommunication Services	4	4	8	6
Pharmaceuticals, Biotechnology & Life Sciences	5	7	3	2
Insurance	6	10	12	12
Financial Services	7	6	7	8
Semiconductors & Semiconductor Equipment	8	8	6	7
Technology Hardware & Equipment	9	5	5	4
Health Care Equipment & Services	10	12	10	10
Household & Personal Products	11	13	13	13
Media & Entertainment	12	9	11	9
Food, Beverage & Tobacco	13	18	18	18
Capital Goods	14	15	15	16
Commercial & Professional Services	15	17	19	17
Utilities	16	16	17	14
Consumer Durables & Apparel	17	19	22	20
Automobiles & Components	18	11	14	24
Materials	19	20	21	22
Consumer Staples Distribution & Retail	20	25	20	19
Energy	21	14	9	11
Consumer Discretionary Distribution & Retail	22	24	25	25
Consumer Services	23	23	24	23
Equity Real Estate Investment Trusts (REITs)	24	21	23	21
Transportation	25	22	16	15

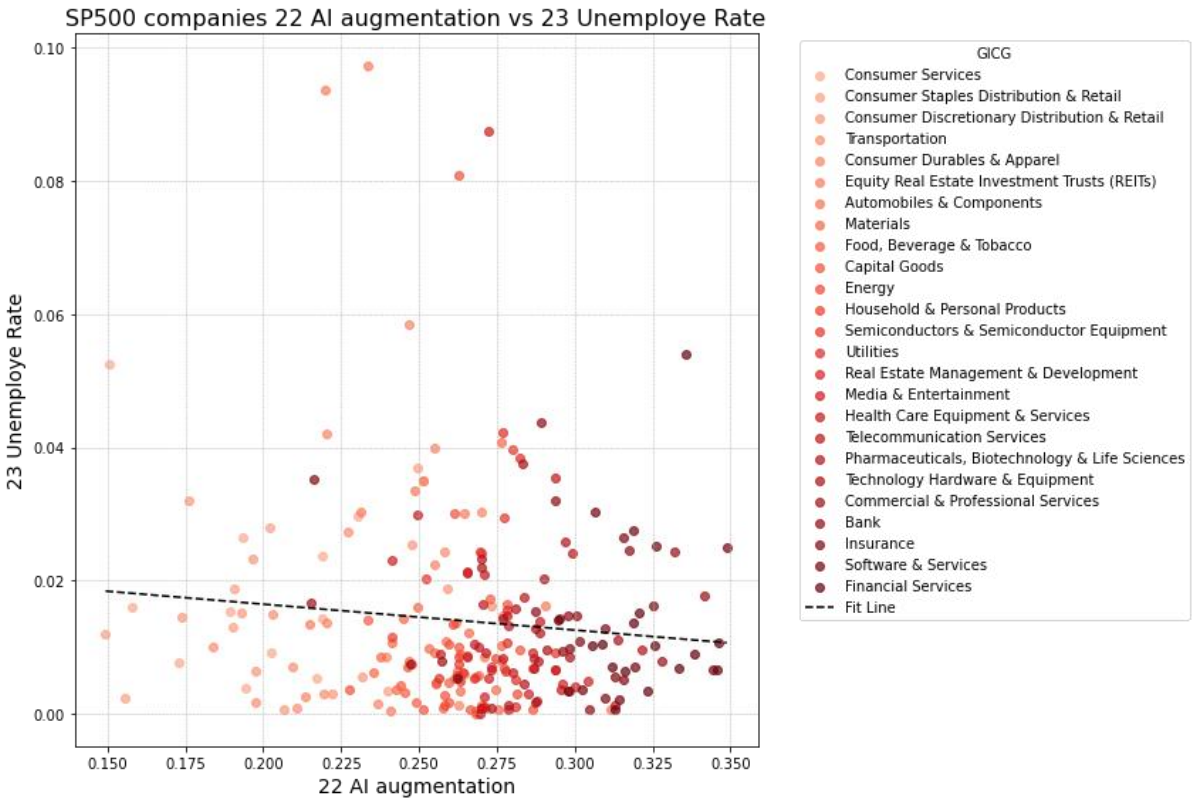
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Table 7: Regression result for all S&P 500 Stocks

	(1)	(2)	(3)	(4)
AI Augmentation$_{job}$	-0.0429***	-0.0384***	-0.0421***	-0.0422***
	(-22.101)	(-19.709)	(-21.885)	(-22.153)
AI Augmentation$_{firm}$		-0.0864***	-0.0748***	-0.068***
		(-24.782)	(-21.714)	(-19.918)
East Coast			0.0700***	0.0662***
			(61.595)	(58.755)
Midwest			0.0665***	0.0626***
			(58.881)	(55.936)
West Coast			0.1129***	0.1076***
			(94.446)	(90.793)
GPT impact				-0.0875***
				(-82.316)
Covid impact				0.0062***
				(6.387)
R-Squared	0.001	0.003	0.028	0.047
Observations	369968	369968	369968	369968

Note: ***, **, * represent statistically significant at the, 1%, 5%, and 10% levels, respectively.

Figure 5: Scatter Plot of AI Augmentation and Layoff Rate Across S&P 500 Companies



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Figure 6: Regression result for all S&P 500 Industries
Note: AI Augmentation is arrayed from left to right, progressing from lowest to highest.

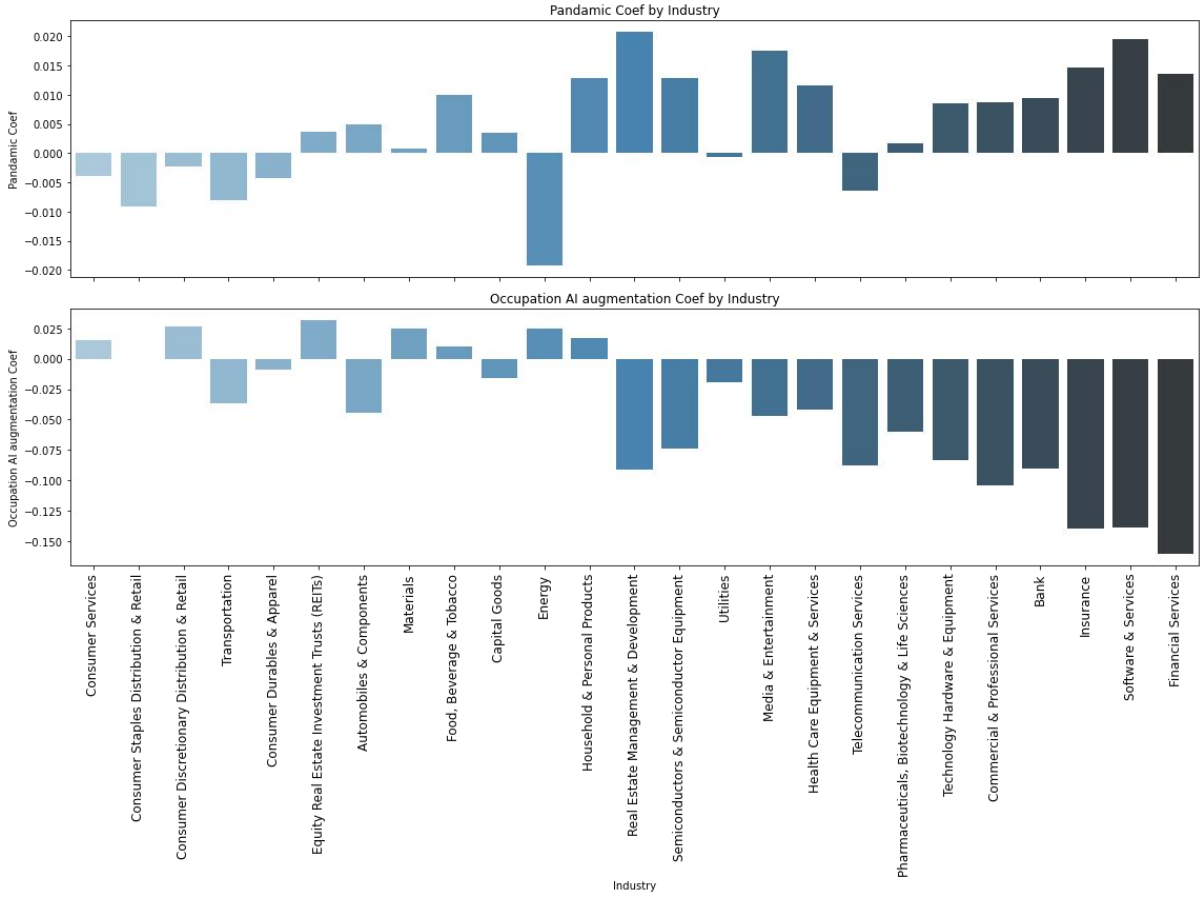
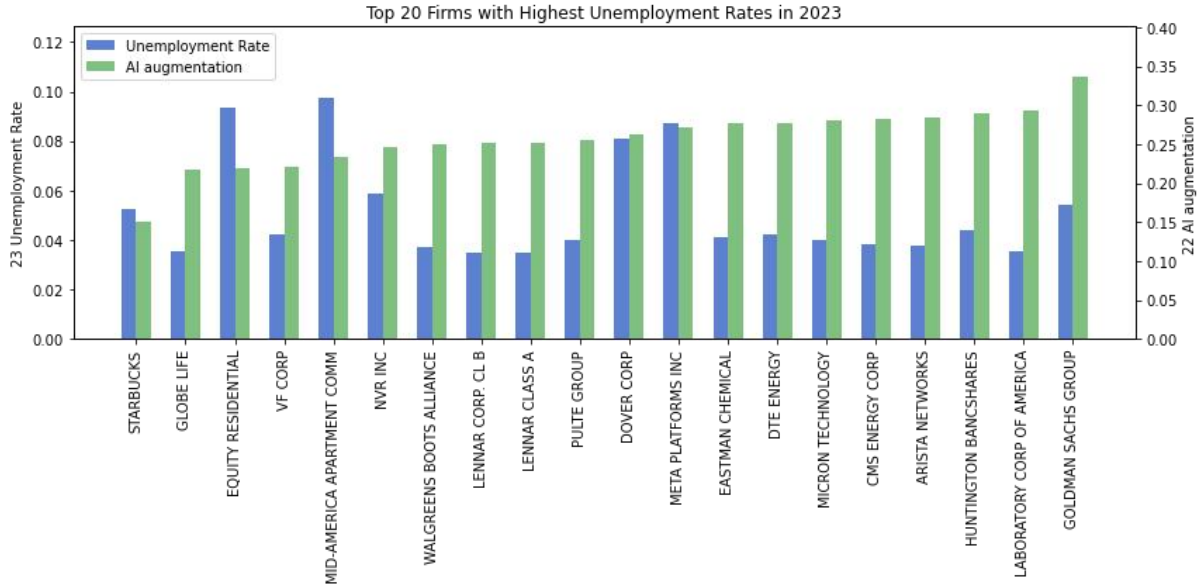


Figure 7: Top 20 Companies with Highest Layoff Rates and their AI Augmentation Levels



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Figure 8: Top 20 Companies with Highest Layoff Number and their AI Augmentation Levels

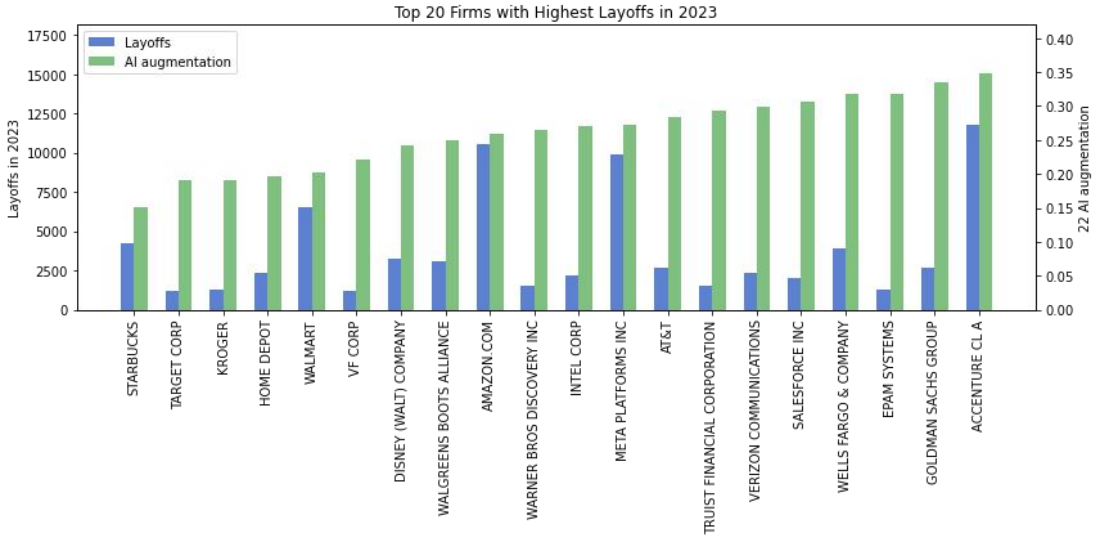


Figure 9: Top 20 Companies with Highest Hire Rates and their AI Augmentation Levels

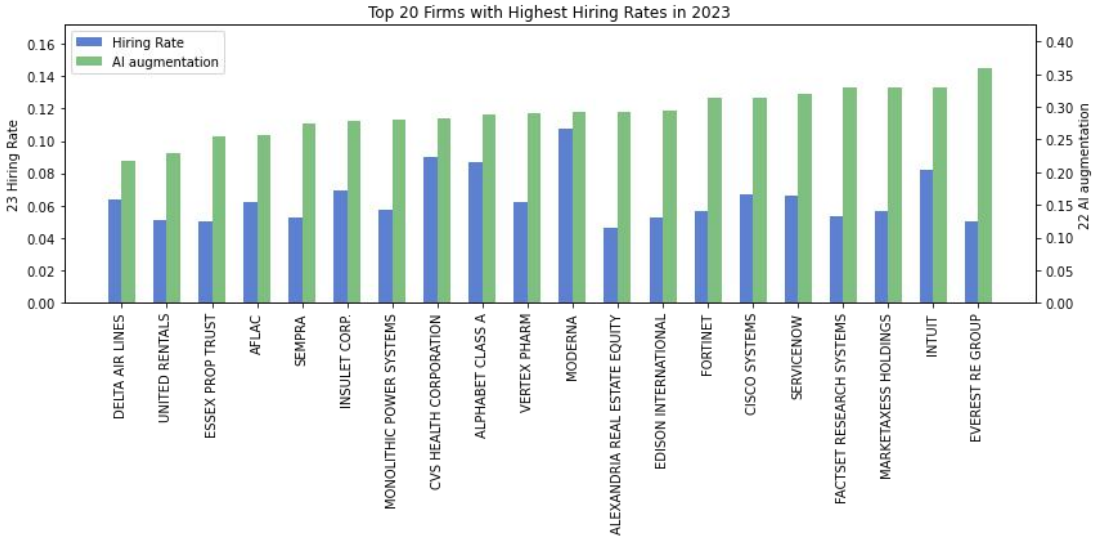
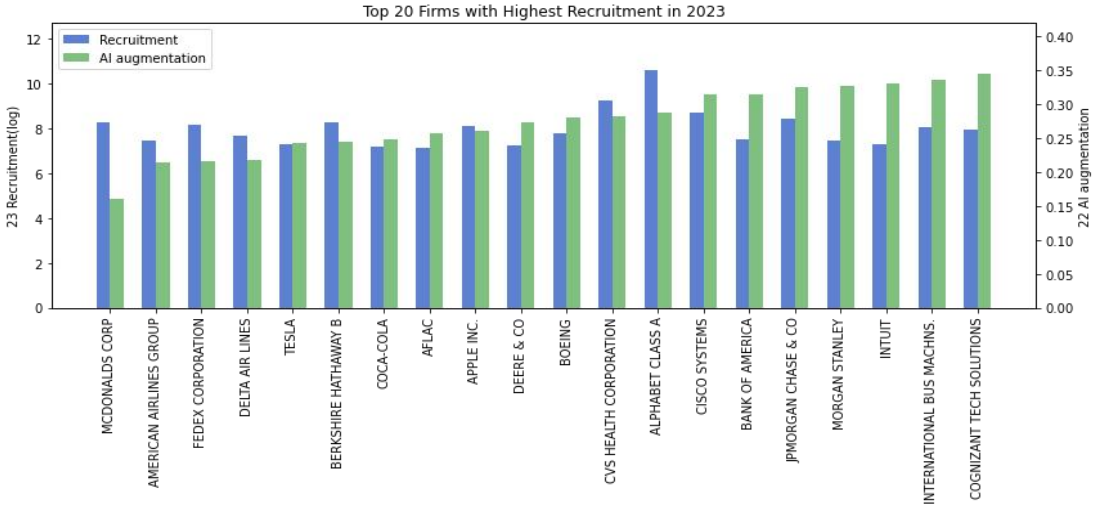


Figure 10: Top 20 Companies with Highest Hire Number and their AI Augmentation Levels



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